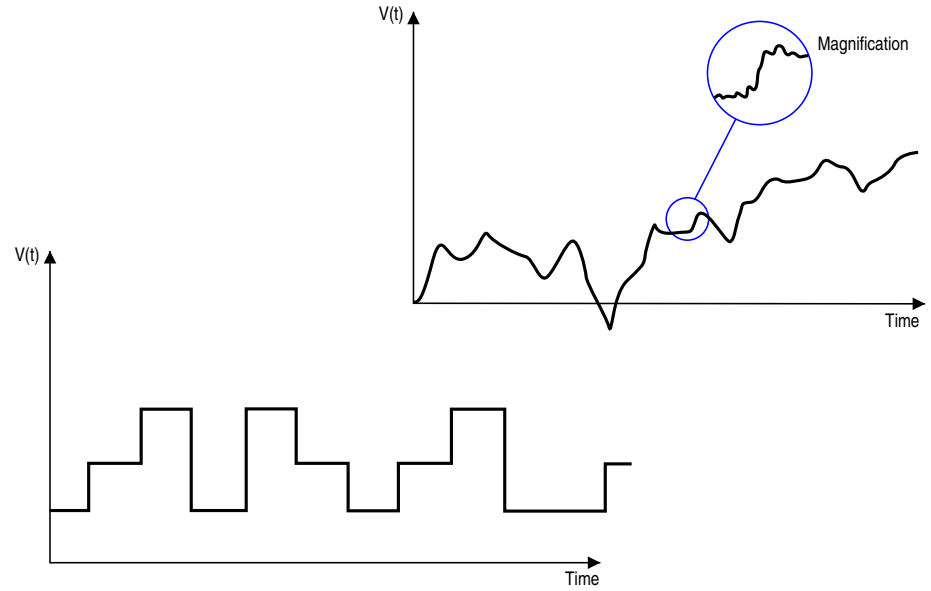
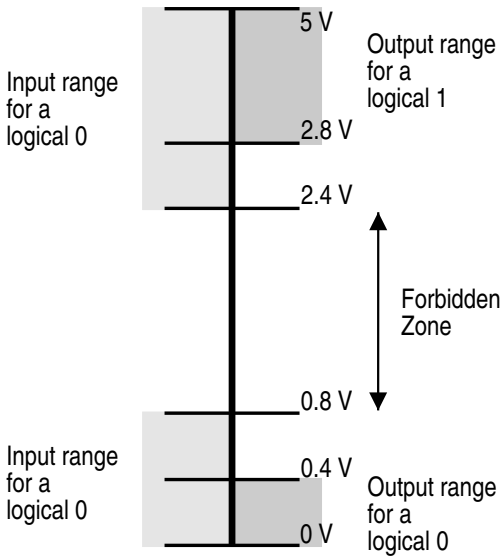
Systems Architecture

Basic Logic Gates

Analog/Digital Signal



Analog to Digital Value



Notes on Digital Values

- Either 1 or 0, no intermediate
- Each value has its inverse (complement)
- A value can be constant or variable
- Values are named:
  START, STOP, RESET, COUNT, ADD, . . .
- START is active-high (1)
  $\overline{\text{START}}$ is active-low (0)

Logic or Algebra

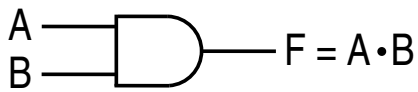
Logic	Boolean Algebra
$A \wedge B$	A • B (AB)
$A \vee B$	A + B
$\neg A$	$\overline{A}$
true	1
false	0
$(A \vee B) \wedge \neg(A \wedge B)$	A $\oplus$ B

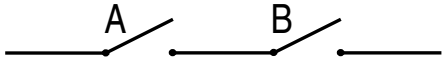
The AND gate

F = A • B

A	B	F
0	0	0
0	1	0
1	0	0
1	1	1

F = A $\wedge$ B



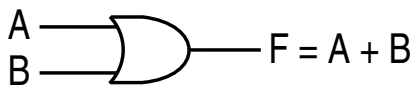


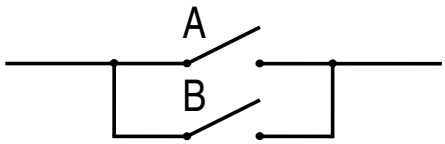
The OR gate

$F = A + B$

A	B	F
0	0	0
0	1	1
1	0	1
1	1	1

$F = A \vee B$



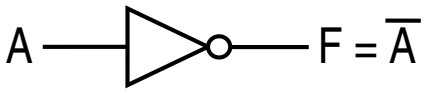


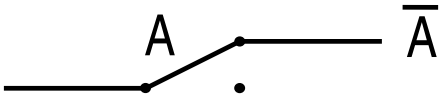
The NOT gate

$F = \overline{A}$

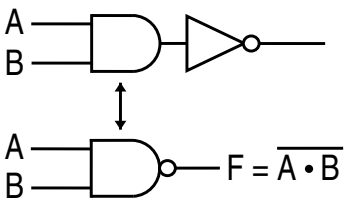
A	F
0	1
1	0

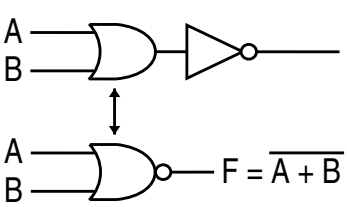
$F = \neg A$





The NAND and NOR gates





$F = \overline{A \cdot B}$

A	B	A • B	$\overline{A \cdot B}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

$F = \neg(A \wedge B)$

$F = \overline{A + B}$

A	B	A + B	$\overline{A + B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

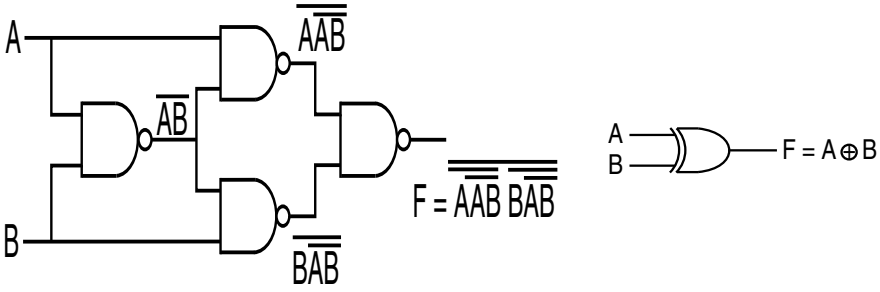
$F = \neg(A \vee B)$

Combinational Logic

- Combination of gates to preform a specific function
- Function expressed in “Boolean Algebra”
- Signal (Propagation) Delay

The Exclusive OR

Known as XOR or EOR



$$F = \overline{(A(\overline{AB}))(B(\overline{AB}))}$$
$$= (A+B)(\overline{AB})$$
$$= A \oplus B$$

Truth Table of XOR circuit

$$\overline{(A(\overline{AB}))(B(\overline{AB}))} \Leftrightarrow (A+B)(\overline{AB}) \Leftrightarrow A \oplus B$$

A	B	$\overline{AB}$	A+B	$(A+B)(\overline{AB})$	$\overline{A(\overline{AB})}$	$\overline{B(\overline{AB})}$	$\overline{(A(\overline{AB}))(B(\overline{AB}))}$
0	0						
0	1						
1	0						
1	1						

$$\neg(\neg(A \wedge (\neg(A \wedge B))) \wedge (\neg(B \wedge (\neg(A \wedge B))))) \Leftrightarrow (A \vee B) \wedge \neg(A \wedge B))$$

A or B but not both